ASSIGNMENT-3

Student Name: Selvadharshini. R

Register Number:732924CSR147

Course Name:BE-CSE

Class:24CS4C

Subject:DATABASE MANAGEMENT SYSTEMS

Assignment Title: SQL Query Practice

Query Execution:

1)create table

```
CREATE TABLE selvas_students(
Studentid INT,
name VARCHAR(20),
```

Department VARCHAR(6),
Marks INT,
City VARCHAR(40)
);

2)insert values into tables

INSERT INTO selvas _students
VALUES(205,'sri','CSE',76,'POOLAMPATTI');
INSERT INTO selvas _students
VALUES(102,'suji','CSE',99,'KONERIPATTI');
INSERT INTO selvas _students
VALUES(123,'neha','ECE',78,'AMMAPETTAI');
INSERT INTO selvas _students
VALUES(110,'selva','EEE',98,'CHENNAI');
INSERT INTO selvas _students VALUES(105,'shalini','CSE',90,NULL);
INSERT INTO selvas _students VALUES(106,'ruba','ECE',99,'ERODE');
INSERT INTO selvas _students
VALUES(109,'reyah','CSE',99,'ERODE');

Sample table:

*	Studentid	name	Department	Marks	City
1	205	sri	CSE	76	POOLAMPATTI
2	102	suji	CSE	99	KONERIPATTI
3	123	neha	ECE	78	AMMAPETTAI
4	110	selva	EEE	98	CHENNAI
5	105	shalini	CSE	90	(null)
6	106	ruba	ECE	99	ERODE
7	109	reyah	CSE	99	ERODE

Queries:

1. **Write an SQL query to display all student records from the Student table**

```
SELECT * FROM selvas _students;
```

*	Studentid	name	Department	Marks	City
1	205	sri	CSE	76	POOLAMPATTI
2	102	suji	CSE	99	KONERIPATTI
3	123	neha	ECE	78	AMMAPETTAI
4	110	selva	EEE	98	CHENNAI
5	105	shalini	CSE	90	(null)
6	106	ruba	ECE	99	ERODE
7	109	reyah	CSE	99	ERODE

2) Write a query to display the Name and Marks of all students

SELECT name, Marks FROM selvas _students;

*	name	Marks
1	sri	76
2	suji	99
3	neha	78
4	selva	98
5	shalini	90
6	ruba	99
7	reyah	99

3) Write a query to display students who scored more than 80 marks

SELECT * FROM selvas _students WHERE Marks > 80;

*	Studentid	name	Department	Marks	City
1	102	suji	CSE	99	KONERIPATTI
2	110	selva	EEE	98	CHENNAI
3	105	shalini	CSE	90	(null)
4	106	ruba	ECE	99	ERODE
5	109	reyah	CSE	99	ERODE

4) Write a query to display students belonging to the CSE department

SELECT * FROM selvas _students WHERE Department='CSE';

*	Studentid	name	Department	Marks	City
1	205	sri	CSE	76	POOLAMPATTI
2	102	suji	CSE	99	KONERIPATTI
3	105	shalini	CSE	90	(null)
4	109	reyah	CSE	99	ERODE

5) Write a query to display students whose city is Chennai

SELECT * FROM selvas _students WHERE City='CHENNAI';

*	Studentid	name	Department	Marks	City
1	110	selva	EEE	98	CHENNAI

6) Write a query to display students whose marks are between 70 and 90

SELECT * FROM selvas _students WHERE Marks BETWEEN 70 AND 90;

*	Studentid	name	Department	Marks	City
1	205	sri	CSE	76	POOLAMPATTI
2	123	neha	ECE	78	AMMAPETTAI
3	105	shalini	CSE	90	(null)

7) Write a query to display student names that end with the letter 'a' using the LIKE operator

SELECT name FROM selvas _students WHERE name LIKE '%a';

*	name
1	neha
2	selva
3	ruba

8) Write a query to display distinct departments from the Student table

SELECT DISTINCT Department FROM selvas _students;

*	Department
1	CSE
2	ECE
3	EEE

9) Write a query to display students whose City value is NULL

SELECT * FROM selvas _students WHERE City IS NULL;

*	Studentid	name	Department	Marks	City
1	105	shalini	CSE	90	(null)

10) Write a query to find the highest marks obtained by a student using an aggregate function

SELECT MAX(Marks) AS HighestMarks FROM selvas _students;

*	HighestMarks
1	99

11) Write a query to find the average marks of students in the ECE department

SELECT AVG(Marks) AS AverageMarks
FROM selvas _students
WHERE Department='ECE';

*	AverageMarks
1	88.5

12) Write a query to count the number of students in each department using GROUP BY

SELECT Department, COUNT(*)
FROM selvas _students
GROUP BY Department;

*	Department	COUNT(*)
1	CSE	4
2	ECE	2
3	EEE	1

13) Write a query to display students whose marks are greater than the average marks of all students (nested subquery).

SELECT * FROM selvas _students
WHERE Marks > (SELECT AVG(Marks) FROM selvas _students);

*	Studentid	name	Department	Marks	City
1	102	suji	CSE	99	KONERIPATTI
2	110	selva	EEE	98	CHENNAI
3	106	ruba	ECE	99	ERODE
4	109	reyah	CSE	99	ERODE

14) . Write an SQL statement to insert a new student record into the Student table.

INSERT INTO selvas _students
VALUES(206,'priya','CSE',90,'SALEM');

New row will be added to existing table:

7	206	priya	CSE	90	SALEM

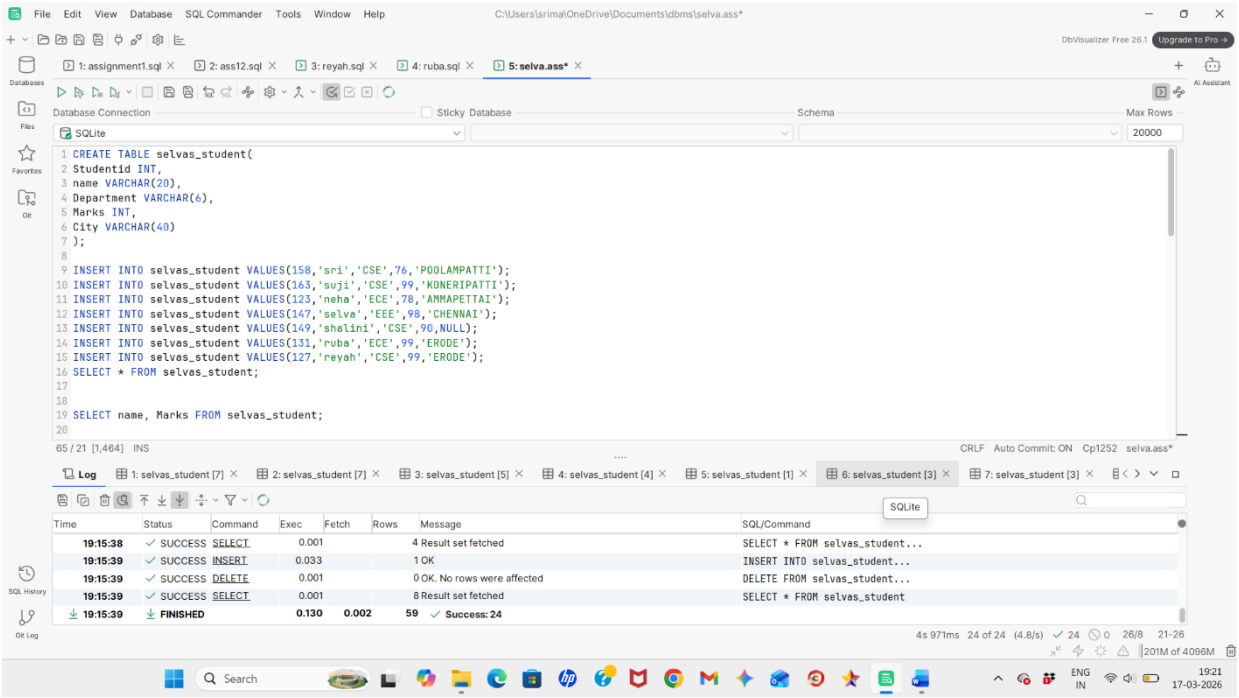
15) Write an SQL query to delete the student with StudentID = 205

DELETE FROM selvas _students
WHERE Studentid=205;

*	Studentid	name	Department	Marks	City
1	102	suji	CSE	99	KONERIPATTI
2	123	neha	ECE	78	AMMAPETTAI
3	110	selva	EEE	98	CHENNAI
4	105	shalini	CSE	90	(null)
5	106	ruba	ECE	99	ERODE
6	109	reyah	CSE	99	ERODE
7	206	priya	CSE	90	SALEM

SOURCE:

DB VISUALIZER



```sql
SELECT * FROM selvas_student WHERE Marks > 80;


SELECT * FROM selvas_student WHERE Department='CSE';


SELECT * FROM selvas_student WHERE City='CHENNAI';


SELECT * FROM selvas_student WHERE Marks BETWEEN 70 AND 90;


SELECT name FROM selvas_student WHERE name LIKE '%a';


SELECT DISTINCT Department FROM selvas_student;

SELECT * FROM selvas_student WHERE City IS NULL;
```

Time	Status	Command	Exec	Fetch	Rows	Message	SQL/Command
19:15:34	SUCCESS	SELECT	0.001		7	Result set fetched	SELECT * FROM selvas_student
19:15:35	SUCCESS	SELECT	0.002		7	Result set fetched	SELECT name, Marks FROM selvas_student
19:15:35	SUCCESS	SELECT	0.001		5	Result set fetched	SELECT * FROM selvas_student WHERE Marks > 80
19:15:35	SUCCESS	SELECT	0.001		4	Result set fetched	SELECT * FROM selvas_student WHERE Department='CSE'
19:15:36	SUCCESS	SELECT	0.001		1	Result set fetched	SELECT * FROM selvas_student WHERE City='CHENNAI'

```sql
SELECT MAX(Marks) AS HighestMarks FROM selvas_student;

SELECT AVG(Marks) AS AverageMarks
FROM selvas_student
WHERE Department='ECE';


SELECT Department, COUNT(*)
FROM selvas_student
GROUP BY Department;

SELECT * FROM selvas_student
WHERE Marks > (SELECT AVG(Marks) FROM selvas_student);


INSERT INTO selvas_student
VALUES(206,'priya','CSE',90,'SALEM');


DELETE FROM selvas_student
```

Time	Status	Command	Exec	Fetch	Rows	Message	SQL/Command
19:15:34	SUCCESS	SELECT	0.001		7	Result set fetched	SELECT * FROM selvas_student
19:15:35	SUCCESS	SELECT	0.002		7	Result set fetched	SELECT name, Marks FROM selvas_student
19:15:35	SUCCESS	SELECT	0.001		5	Result set fetched	SELECT * FROM selvas_student WHERE Marks > 80
19:15:35	SUCCESS	SELECT	0.001		4	Result set fetched	SELECT * FROM selvas_student WHERE Department='CSE'
19:15:36	SUCCESS	SELECT	0.001		1	Result set fetched	SELECT * FROM selvas_student WHERE City='CHENNAI'

```sql
WHERE Department='ECE';


SELECT Department, COUNT(*)
FROM selvas_student
GROUP BY Department;

SELECT * FROM selvas_student
WHERE Marks > (SELECT AVG(Marks) FROM selvas_student);


INSERT INTO selvas_student
VALUES(206,'priya','CSE',90,'SALEM');


DELETE FROM selvas_student
WHERE Studentid=205;


SELECT * FROM selvas_student;
```

Time	Status	Command	Exec	Fetch	Rows	Message	SQL/Command
19:15:37	SUCCESS	SELECT	0.001		3	Result set fetched	SELECT DISTINCT Department FROM selvas_student
19:15:37	SUCCESS	SELECT	0.001		1	Result set fetched	SELECT * FROM selvas_student WHERE City IS NULL
19:15:37	SUCCESS	SELECT	0.001		1	Result set fetched	SELECT MAX(Marks) AS HighestMarks FROM selvas_student
19:15:38	SUCCESS	SELECT	0.002	0.001	1	Result set fetched	SELECT AVG(Marks) AS AverageMarks...